

Unique ID: 16590

Lecture : CPE 2.220, Tu and Th, 11 AM- 12:30 PM

Recitation: Mandatory. Aspen software sessions will be in recitation only. Wed 4-6 RLP 0.128

Required Text: McCabe, Smith and Harriott "Unit Operations of Chemical Engineering" Sixth or Seventh edition (These editions are very similar). Electronic rental of the book is less expensive and just fine.

Related references in chemistry library:

Cussler "Diffusion: Mass Transfer in Fluid Systems " (non-required suppl. text – key info in sup. Notes in Canvas, we will use this widely as well).

Prof: Manish Kumar, Office Hours: CPE 5.162 T and Th 4:30-5:30
manish.kumar@utexas.edu

TA: RJ Vogler, Office Hours: T 3-4 and Th 2-3 in CPE 2.222
rjv684@utexas.edu

Prerequisites: ChE 322 Thermo, ChE 319 Transport

Chapters (McCabe et al. (6th or 7th edition required- very similar)

UNIT 1:

Chapter 17 Diffusion and Mass Transfer between Phases (steady and unsteady state) (also Cussler) ~3 ½ weeks

First Exam

UNIT 2:

Chapter 18 Gas Absorption (excluding rich gases and chemical reaction) 1.5

Chapter 20 Equilibrium stage operations 1/2

Chapter 21 Distillation 2

Second Exam

UNIT 3:

Chapter 25 Adsorption 2

Chapter 26 Membrane separations; gases, liquids- osmosis and reverse osmosis 2

Third Exam

Final Exam (Cumulative)

Grading method: 3 Hourly exams (total of 60%), Final exam (25%), HW (10%) Aspen Project (5%).

Exams

- Mid-term Unit Exam dates: Feb 12th , March 12th , April 16th ,
- Finals : Saturday, May 2nd , 8 AM – 10 AM, Closed book, Mass transfer summary sheet and all needed equations provided. No notes.

Details:

- In class exams (~1.5 hr)
- Questions for each unit exam will be directly from a ~20 question bank and could include questions from HW or in class problems, ~80% of the grade from these questions. Some additional sections will be added to these questions (for example an additional sub question for a main problem). Mass transfer summary sheet and all needed equations provided.
- Final exams will include all question banks, HWs, and in class problems.

Homework: Homework will be assigned officially under assignments in Canvas for most weeks to encourage continuous learning. Upload into assignments on Canvas. HWs will be due on Thursday at 11:59 PM

Lecture start time: Pls arrive a few minutes early so you are ready to go.

Coordination of primary power point notes, supplemental notes, and text: Pls see documents on Canvas in Files under various chapters.

Canvas: “Files” includes general info as well as material specific to each chapter as filed under the folder Chapters.

LearnChemE screencasts: <http://www.learncheme.com>. See separations/mass transfer screencasts All of you learn differently. We wish to reach everyone. These highly acclaimed optional screencasts for PCs to tablets to phones may be very helpful to many of you.

Abilities before entering

Understanding of vapor-liquid equilibrium and liquid-liquid eq. Mass and energy balances, Fick's laws and mass transfer section of ChE 319.

Use of spread sheets which will be used to understand solutions of stage wise problems.

Abilities students should gain

Outcome: Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

Engineering design for open ended problems: How to synthesize material balances, thermodynamics and mass transfer to do stagewise calculations- often requires engineering assumptions

Solution of stage wise mass transfer and equilibrium problems
spreadsheets- to learn concepts of McCabe-Thiele graphical solutions
process simulator (e.g. Aspen)

Thermodynamic calculations in distillation and stage efficiencies

Understand key components and concentration profiles in multicomponent distillation

Mass transfer coefficients, correlations and film theory (absorption, adsorption etc.)

Partial differential equations: mass transfer and adsorption chapters

Molecular concepts and materials chemistry for achieving selectivity in separations

Nucleation, growth and crystallization (examples given for nanotechnology)

Consideration of safety, energy utilization, sustainable technologies, pollution, emerging technologies and the role of government policy and society

Application to broad range of industries: energy, petrochemicals, environmental, nanotechnology, biotechnology, pharmaceuticals-drug delivery, transportation (fuel cells)

Tips to maximize success:

1. It is an honor to have you in our department and in my class as you have already excelled. Continue to develop a passion for knowledge and life-long learning!
2. Class etiquette and professionalism- maximize learning efficiency by giving the course your full attention during class.
3. Review info where you first learned it. Reread key sections in calculus book, transport book, thermo book etc. and your old notes. The more you review these topics the better you will retain it.
4. The textbook is a key element for this class. As you do lifelong learning, reading books and articles play an important role. **The notes are not meant to be self-sufficient, but to complement the text. Read them together!! The notes refer to text locations.**
5. See us in office hours. Participate in lecture and recitation. Students often chat with me right after class as well.
6. Find other students that study together online.
7. **It usually takes ~ 8 hours/week outside of class to read the book including examples in chapters (in conjunction with notes), learn the theory and concepts, and to do the homework problems.**
8. Strive to understand the theory and concepts in addition to doing problems.
9. Learn to identify classes of problems based on theoretical concepts. The summary sheets called exam study guide in the chapters will help here.
10. Maximize what you get out of the lectures by keeping up and understanding all concepts after each lecture. If you get behind, try to immediately catch up to max. efficiency.
11. Look ahead in the book and notes for about 5-10 min before each lecture. The brain will then be activated and eager to learn very efficiently and deeply.
12. Avoid the short cut to use old homework solutions as a crutch. It is far easier to follow somebody else's solution than to construct your own.
13. Students should do all homework problems and study solutions after turning in homework as needed on Canvas. Save hws when you get them back so you can study your approach vs. posted solutions.
14. Always check units in eqns. They are important. Check with factor label. This approach helps with exam grades.

- Academic Integrity statement.

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.

- Confidentiality of Class Recordings.

Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

- University Health Services.

Important safety information and University health services resources can be found here: <https://www.healthyhorns.utexas.edu/index.html>

- Disability & Access (D&A).

The university is committed to creating an accessible learning environment consistent with university policy and federal and state law. Please let your instructors know if you experience any barriers to learning so they can work with you to ensure you have equal opportunity to participate fully in your courses. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability & Access (D&A). Please refer to the [D&A website](#) for more information. If you are already registered with D&A, please deliver your accommodation letter to your instructors as early as possible in the semester so you can discuss together your approved accommodations and needs in your courses.

- Course Instructor Survey/Evaluation.

An evaluation of the course and instructor will be conducted at the end of the semester using the approved UT Course/Instructor evaluation forms.

- Undergraduate student Drop policy.

From the 1st through the 12th class day (4th class day in the summer sessions), an undergraduate student may drop a class through the registration system. From the 13th (5th class day in the summer sessions) through the university's academic drop deadline, a student may Q drop a class with approval from the Dean, and departmental advisor.

- Emergency Preparedness Plan.

Emergency Preparedness means being ready. It takes an effort by all of us to create and sustain an effective emergency preparedness system. You are your own best first responder. Please use <https://preparedness.utexas.edu/welcome-emergency-preparedness> as a resource to better understand emergency preparedness at the university, and how you can become part of and contribute to the preparedness community. To monitor emergency communications for specific instructions, go to <https://utexas.edu/emergency>. To report an issue (non emergency) call 512-471-4441. In case of emergency, call 911.

- **“All other university policies not explicitly included on this syllabus can be found on the General Information Catalog: <http://catalog.utexas.edu/general-information/>.”**
- A notice regarding **Academic Dishonesty, UT Honor Code** (or statement of ethics), and an explanation of what constitutes plagiarism. (*Note: Student Judicial Services implements and interprets University regulations related to student behavior and works with faculty members regarding academic dishonesty.*) **For additional information, see the Student Conduct and Academic Integrity or SCAI website and University General Information Catalog at: <http://deanofstudents.utexas.edu/conduct/> and <https://catalog.utexas.edu/general-information/appendices/appendix-c/student-conduct-and-academic-integrity/>**

- **ARTIFICIAL INTELLIGENCE**

“The use of generative artificial intelligence tools (or Large Language Models [LLMs]) such as CoPilot or ChatGPT in this class shall be permitted on a limited basis for learning material but will not be accepted if used for competing assignments. You are also welcome to seek my prior-approval to use generative AI on any assignment if it is unclear as to when and how it can be used. In either instance, generative AI should be used with caution and proper citation, as the use of generative AI must be properly attributed. Using generative AI without authorization, or failing to cite generative AI use according to the citation policy in this course, even where permitted, may constitute a violation of UT Austin’s Institutional Rules on academic integrity and may be referred to the Office of Student Conduct and Academic Integrity for resolution.

- Accommodations for **religious holy days**. The following statement is from the University General Info. Catalog, <http://catalog.utexas.edu/general-information/academic-policies-and-procedures/attendance/>.

“A student who misses classes or other required activities, including examinations, for the observance of a religious holy day should inform the instructor as far in advance of the absence as possible, so that arrangements can be made to complete an assignment within a reasonable time after the absence.”

- **The following recommendations regarding emergency evacuation from the Office of Campus Safety and Security, 512-471-5767, <https://www.utexas.edu/campus-life/safety-and-security>:**
 - Occupants of buildings on The University of Texas at Austin campus are required to evacuate buildings when a fire alarm is activated. Alarm activation or announcement requires exiting and assembling outside (across the bridge).
 - Familiarize yourself with all exit doors of each classroom and building you may occupy. Remember that the nearest exit door may not be the one you used when entering the building.
 - Students requiring assistance in evacuation shall inform their instructor in writing during the first week of class.
 - In the event of an evacuation, follow the instruction of faculty or class instructors. Do not re-enter a building unless given instructions by the following: Austin Fire Dept., The University of Texas at Austin Police Dept., or Fire Prevention Services office.
 - **Behavior Concerns Advice Line (BCAL) 512-232-5050. For more information visit the BCAL website: <https://safety.utexas.edu/behavior-concerns-advice-line>**
 - Link to information regarding emergency evacuation routes and emergency procedures can be found at: www.utexas.edu/emergency

- **TITLE IX DISCLOSURE**

Texas Education Code, Section 51.252 requires all employees of Texas universities, including faculty, to report to the [Title IX Office](#) any information regarding incidents of sexual harassment, sexual assault, dating violence, or stalking that is disclosed to them. Texas law requires that all employees who witness or receive information about incidents of this nature (including, but not limited to, written forms, applications, one-on-one conversations, class assignments, class discussions, or third-party reports) must report it to the Title IX Coordinator. Before talking with any faculty or staff member about an incident of this nature, please be aware that they are required to report that information.

Graduate teaching and research assistants are also designated as [mandatory reporters](#) and are required to report [behaviors that are considered sexual misconduct](#), including the types of misconduct described above.

Pregnant and parenting students are also eligible to receive support and resources from the Student Parent Liaison in the [Title IX Office](#). Support and resources may include supportive measures, medically necessary periods of leave, and leaves of absence. Case managers from the Title IX Office connect with students to provide one-on-one, individualized support. If you would like to speak with a case manager for support, please email: supportandresources@austin.utexas.edu.

For more information about reporting options and resources, please visit the [Title IX](#) website, contact the Title IX Office via email at: titleix@austin.utexas.edu, or call 512-471-0419.